Spartan

1. IDE

We strongly suggest use cursor (https://cursor.com) to help you fix or write code via the help of AI. You should have a pro version using your university email: https://cursor.com/students. Find the word: verify status.

2. Config ssh

Install Remote ssh extension. And config your Spartan account. You can either use passwd or use your key to avoid input passwd https://dashboard.hpc.unimelb.edu.au/ssh/

```
Host spartan
    HostName spartan.hpc.unimelb.edu.au
    User Your_Unimelb_Userame
    PreferredAuthentications publickey
    IdentityFile ~/.ssh/id_rsa
```

3. Connect to spartan and terminal (you do not need VPN)

Connect to spartan via remote ssh extenstion (click bottom left). Then use Cursor's default terminal. Once you connected to spartan your terminal will locate you into spartan server automatically.

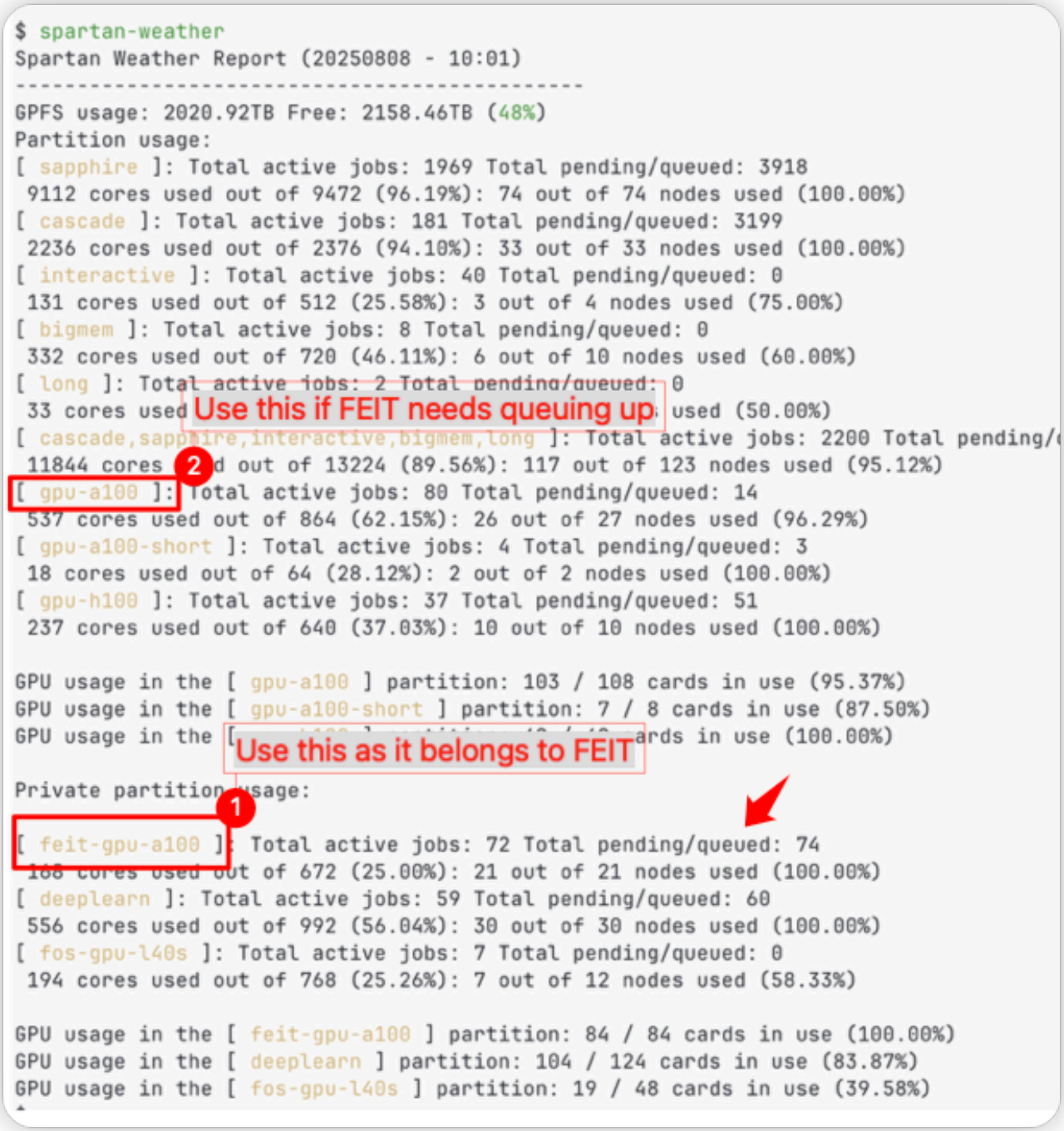
You MUST to work/install everything under your project folder `/data/gpfs/projects/punimxxxx/yourusername/`. Install miniconda Installing Miniconda or mamba or uv etc... and Make sure your miniconda and your data in NOT in your home directory i.e. `/home/Your_Unimelb_Username`.

4. Use interactive mode

You should use interactive mode to debug your code. In this case you will automatically get gpu to run your python code by `python xxx.py`. If you want 30 minutes to debug run:
`sinteractive -p interactive --time=00:30:00 --cpus-per-task=8 --partition=gpu-a100-short --nodes=1 --gres=gpu:1`
Change the time depend on how long you want to debug using the gpu.

5. Submit a task

If you think your code is ready you can submit a task to run (enjoy your coffee or paper writing :)).
First find out which gpu you want to submit: `spartan-weather`
You will see:



Then do: `vim your_task_name.slurm`

write:

```
#!/bin/bash
#SBATCH --partition=feit-gpu-a100
#SBATCH --account=punim2243
#SBATCH --qos=feit
#SBATCH --nodes=1
#SBATCH --job-name="fla-tts"
#SBATCH -o "slurm-%N.%j.out" # STDOUT
#SBATCH -e "slurm-%N.%j.err" # STDERR
```

```
#SBATCH --ntasks=1
#SBATCH --gres=gpu:1
#SBATCH --mail-user=yourusername@unimelb.edu.au
#SBATCH --mail-type=FAIL
#SBATCH --time=0-12:00:00
#SBATCH --mem=64G
# Send yourself an email when the job:
# aborts abnormally (fails)
#SBATCH --mail-type=FAIL
# begins
#SBATCH --mail-type=BEGIN
# ends successfully
#SBATCH --mail-type=END

cd /data/gpfs/projects/project_name/your_project || exit 1 # Exit if cd fails

module purge
# Load CUDA version compatible with the cuDNN module
module load CUDA/12.4.1
module load cuDNN/9.6.0.74-CUDA-12.4.1

# Activate the bash virtual environment
conda activate your_environment
# run the code
python yourscript.py
```

You MUST need to 1. cd to your *.py folder 2. activate your env
before submit slurm.

If you everything is ready. Then:
`sbatch your_task_name.slurm`
you can check your submitted task ID via:
`squeue --me`
And if you want to stop:
`scancle task_ID`

Enjoy!